NATIONAL JUNIOR COLLEGE
SENIOR HIGH 2
Preliminary Examination

NAME	

SUBJECT CLASS	2ma2	REGISTRATION NUMBER	

H2 MATHEMATICS 9758 / 02

17 September 2019

3 hours

Candidates answer on the Question Paper.

Additional Materials: List of Formulae (MF26)
Writing Paper (1 sheet)

READ THESE INSTRUCTIONS FIRST

Write your name, class and registration number in the boxes above.
Please write clearly and use capital letters.

Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use paper clips, glue or correction fluid.

Answer **all** the questions.

Write your answers in the spaces provided in the question paper.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

The use of an approved graphing calculator is expected, where appropriate.

Unsupported answers from a graphing calculator are allowed unless a question specifically states otherwise.

Where unsupported answers from a graphing calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.

You are reminded of the need for clear presentation in your answers. Up to 2 marks may be deducted for improper presentation.

The number of marks is given in the brackets [ ] at the end of each question or part question.

Question Number	Marks Possible	Marks Obtained
1	8	
2	9	
3	10	
4	13	
5	5	
6	6	
7	7	
8	9	
9	9	
10	12	
11	12	
Presentation Deduction		–1 / –2
TOTAL	100	

[Turn over

Section A: Pure Mathematics [40 marks]

1 It is given that

$$f(x)=\begin{cases} ax, & \text{for } 0 \le x < 2, \\ \dfrac{12a}{x}-4a, & \text{for } 2 \le x < 3, \end{cases}$$

where a is a positive constant and that $f(x)=f(x+3)$ for all real values of x.

(i) Find the exact value of $f\left(\dfrac{44}{3}\right)$ in terms of a. [2]

(ii) Sketch the graph of $y=f(x)$ for $-3 \le x \le 3$, indicating clearly the coordinates of the maximum points and axial intercepts. [3]

(iii) Find $\int_{-2}^{2} f(x)\,dx$ in terms of a, leaving your answer in the exact form. [3]

2 Find the general solution of the differential equation

$$\frac{d^2y}{dx^2}=\frac{2}{\sqrt{1-4x^2}}, \text{ where } -\frac{1}{2}<x<\frac{1}{2}. \qquad [5]$$

The particular solution curve $y=f(x)$ has a minimum point at the origin.

(i) Find $f(x)$. [2]

(ii) Sketch the graph of this particular solution. [2]

3 **Do not use a calculator in answering this question.**

(a) **(i)** Given that $x+y\mathrm{i}=\sqrt{8-6\mathrm{i}}$, determine the possible values of x and y, where x and y are real numbers. [2]

(ii) Solve $w^2+2\mathrm{i}w-9+6\mathrm{i}=0$, giving your answer in the form $a+b\mathrm{i}$, where a and b are real numbers. [2]

(b) **(i)** Show that if z_1 is a root of the equation $z^4-pz^2+q=0$, where p and q are constants, then $-z_1$ is a also a root of this equation. [1]

(ii) Verify that $3e^{\mathrm{i}\left(\frac{\pi}{6}\right)}$ is a root of the equation $z^4-9z^2+81=0$. Hence express z^4-9z^2+81 as the product of two quadratic expressions in z with exact real coefficients. [5]

4 If k is a nonzero integer, find $\int_{-\pi}^{\pi} \sin kx \, dx$ and $\int_{-\pi}^{\pi} \cos kx \, dx$. [2]

Let m and n be positive integers.

(i) Show that

$$\int_{-\pi}^{\pi} \sin(mx)\sin(nx)\,dx = \begin{cases} \pi & \text{if } m = n, \\ 0 & \text{if } m \neq n. \end{cases}$$ [4]

(ii) By considering the formula for $\cos(mx+nx)$, evaluate

$$\int_{-\pi}^{\pi} \cos(mx)\cos(nx)\,dx.$$ [2]

Suppose a is a positive integer.

(iii) Show that $\int_{-\pi}^{\pi} |\sin(ax)|\,dx = 4$. [2]

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Hence, find the exact volume of the solid formed when the region bounded by the curve $y = |\sin(ax)| + 1$, the x-axis and the lines $x = -\pi$ and $x = \pi$, is rotated completely about the x-axis. [3]

Section B: Probability and Statistics [60 marks]

5 The random variable X is the number of successes in n independent trials of an experiment in which the probability of a success at any one trial is p.

(i) Show that $\dfrac{P(X=k+1)}{P(X=k)} = \dfrac{(n-k)p}{(k+1)(1-p)}$, where $k = 0, 1, 2, \ldots, (n-1)$. [2]

(ii) Using the result in part **(i)**, find the most probable number of successes when $n = 99$ and $p = \dfrac{7}{9}$. [3]

6 A group of 8 people consists of 4 married couples.

(a) The group stands in a line. Find the number of different possible orders in which no two men stand next to each other. [2]

(b) The group stands in a circle. Find the number of different possible orders in which each man stands next to his wife. [2]

(c) The group forms a committee consisting of two teams of four people each. Find the number of ways that the committee can be formed such that neither team consists of only men or women. [2]

7 A survey was conducted with a large number of employees who were asked to select one SkillsFuture programme that they would like to enroll in to further develop their skills. The survey showed that 29% would like to enroll in human resource programme, 100p% would like to enroll in financial management programme and the rest would like to enroll in infocomm technology programme.

Fifteen employees who took part in the survey were randomly selected. Find the probability that

(i) exactly twelve employees said that they would like to enroll in either infocomm technology programme or financial programme, [2]

(ii) no less than five employees said that they would like to enroll in human resource programme. [2]

Suppose now that four employees who took part in the survey are randomly selected, write down an equation in terms of p such that the probability of one or two employees saying that they would like to enroll in financial management programme is 0.3. Hence find the value of p. [3]

8 Joe has won a game. For his prize, he is given k identical sealed envelopes out of which he is allowed to open exactly three and keep their contents. Three of the envelopes each contain \$1 and the rest each contain \$10. He chooses three envelopes at random. Let \$$X$ be the amount of prize money that he receives when he wins the game once.

The probability distribution of X is given in the table below

x	3	12	21	a
$\mathrm{P}(X=x)$	0.05	0.45	0.45	0.05

(i) Find the values of a and k. [2]

(ii) Explain the concept of randomness in the context of this question. [1]

Joe won the game n times where $n \geq 20$.

(iii) Find the least value of n such that the probability that the mean amount of prize money that Joe receives is within \$2 of the expected amount of money that he will receive after winning one game, is greater than 0.95. [6]

9 *A temperature anomaly* is the change in global surface temperature relative to the average temperature from Year 1951 to 1980.

The table gives the temperature anomaly in degree celsius from Year 2011 to 2018.

t, number of years since 1980	31	32	33	34	35	36	37	38
Temperature Anomaly, θ°C	0.57	0.61	0.64	0.73	0.86	0.98	0.90	0.82

source: https://climate.nasa.gov

(i) Draw a scatter diagram for these values, labelling the axes. [2]

(ii) Calculate the value of the product moment correlation coefficient for this set of data. [1]

(iii) State, giving a reason, which of the least squares regression lines, θ on t or t on θ, should be used to express a possible linear relation between θ and t. [1]

(iv) Calculate the equation of the regression line chosen in part **(iii)**. Hence, interpret, in context, the value of the gradient of the regression line. [2]

(v) Use the regression line to predict the temperature anomaly for Year 2025. Comment on the reliability of this estimate. [2]

(vi) Explain why in this context a linear model would probably not be suitable for long term predictions. [1]

10 To determine if a patient has heart disease, a test called *cardiac fluoroscopy* is used. This test checks for narrowing of heart arteries due to build-up of calcium. Suppose that a *cardiac fluoroscopy* test gives either a positive or negative result. Let

T_0 be the event of obtaining a negative test result indicating that no artery is narrowing,
T_1 be the event of obtaining a positive test result indicating that at least one artery is narrowing,
S be the event that heart disease is present in the patient, and
S' be the event that heart disease is not present in the patient.

Based on studies, the conditional probabilities are given in the following table.

i	$\mathrm{P}(T_i \mid S)$	$\mathrm{P}(T_i \mid S')$
0	0.42	q_1
1	q_2	0.04

(i) Find q_1 and q_2. [2]

Suppose $\mathrm{P}(S) = p$.

(ii) Show that $\mathrm{P}(S \mid T_0) = \dfrac{7p}{16 - 9p}$. [3]

(iii) Prove by differentiation that $\mathrm{P}(S \mid T_0)$ is an increasing function for $0 < p < 1$, and explain what this statement means in the context of the question. [3]

(iv) It is further given that $p = 0.3$ for a group of patients. One randomly chosen patient from this group takes the *cardiac fluoroscopy* test twice and both test results are positive. Find the probability that the patient has heart disease. You may assume the test results are independent of each other. [4]

11 In view of the rise in the number of accidents involving Personal Mobility Devices (PMDs) in the past year, regulations on PMDs are being tightened to step up safety for pedestrians on footpaths. Since February 2019, the speed limit for riding a PMD on footpaths has been lowered to 10 km/h. In June 2019, a concerned citizen believes that the mean speed of PMD riders still exceeds the new speed limit. To test his belief, a team of enforcement officers capture the speeds, x km/h, of a random sample of 250 PMD riders at five hotspots. The findings are summarised by

$$\sum(x-10)=38.9, \qquad \sum(x-10)^2=390.31.$$

Assume that the speed of a PMD rider follows a normal distribution.

(i) Calculate unbiased estimates of the population mean and variance of the speed of PMD riders on footpaths. [2]

(ii) Test at the 4% level of significance whether the belief should be accepted. [4]

(iii) Explain, in the context of the question, the meaning of ‘at the 4% level of significance’. [1]

(iv) Without any further test, explain if the conclusion in part **(ii)** will change if instead we are testing that the mean speed of PMD riders on footpaths is not 10 km/h. [1]

Three months later, to investigate if the mean speed of PMD riders on footpaths differs from the regulated speed limit, another large random sample is collected at the same hotspots. It is found that the mean speed captured is 9.5 km/h, with standard deviation 3.2 km/h.

(v) Find the smallest sample size that will provide significant evidence at the 5% level of significance that the mean speed of PMD riders on footpaths differs from the regulated speed limit. [4]